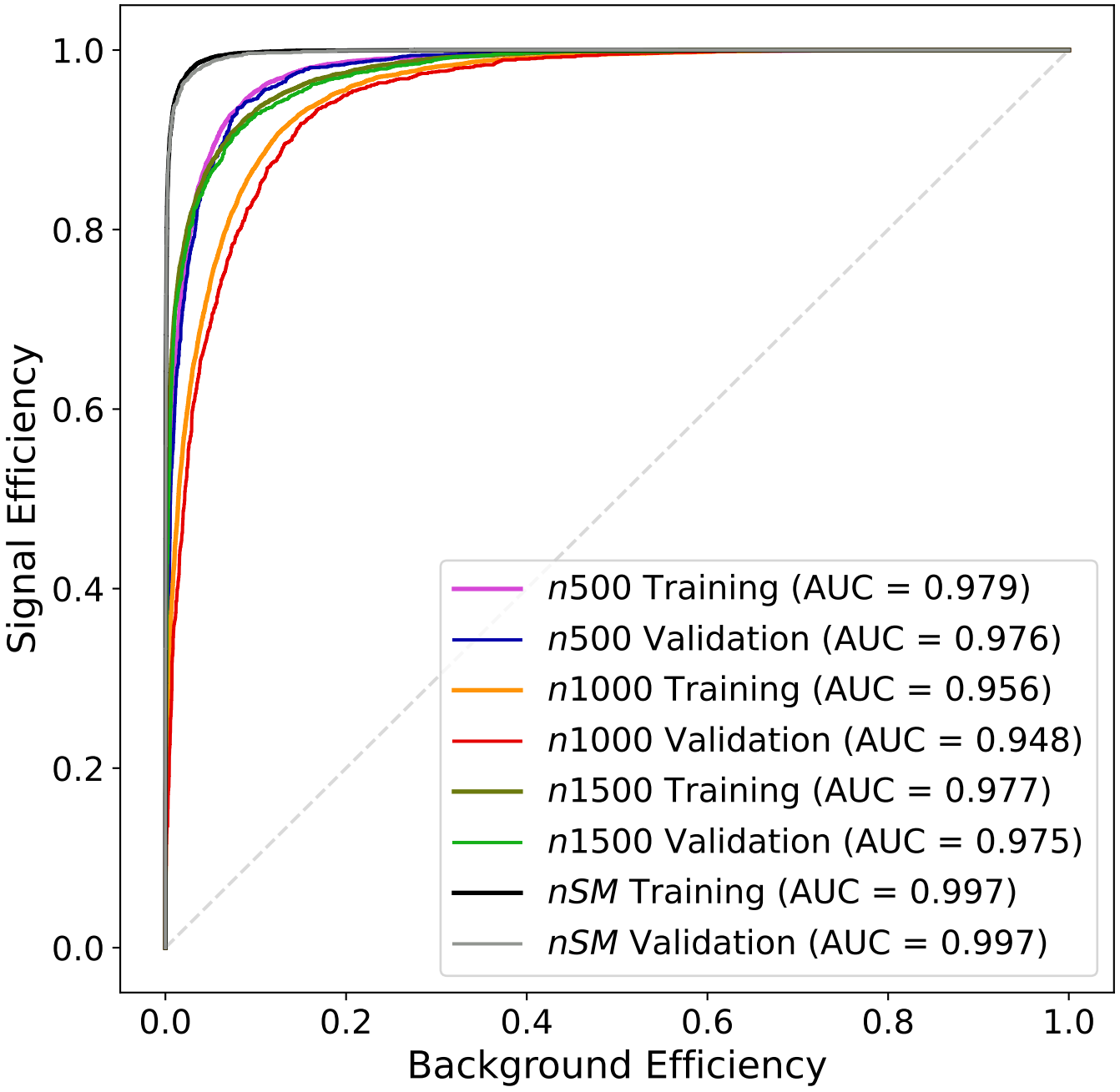
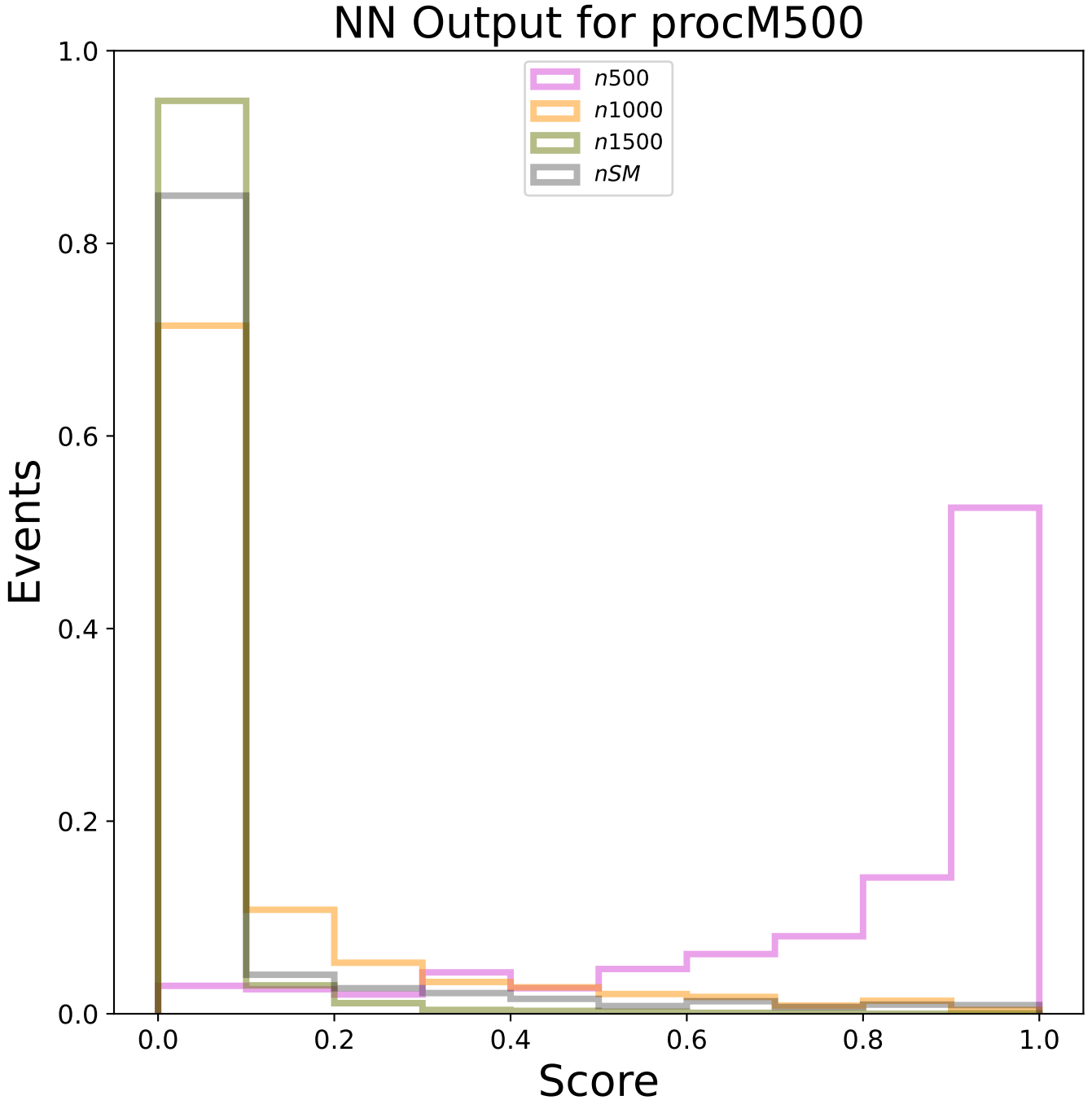
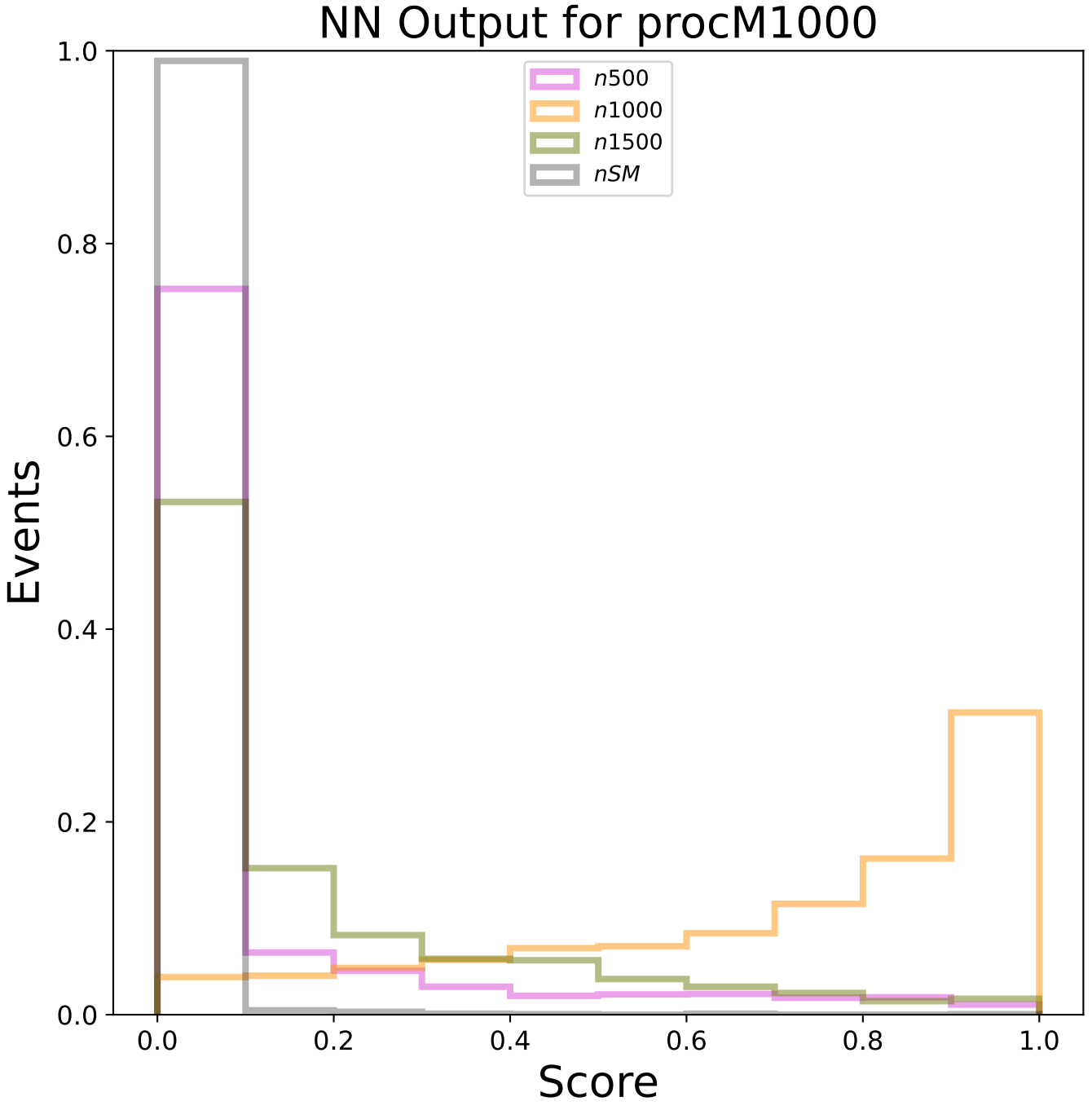
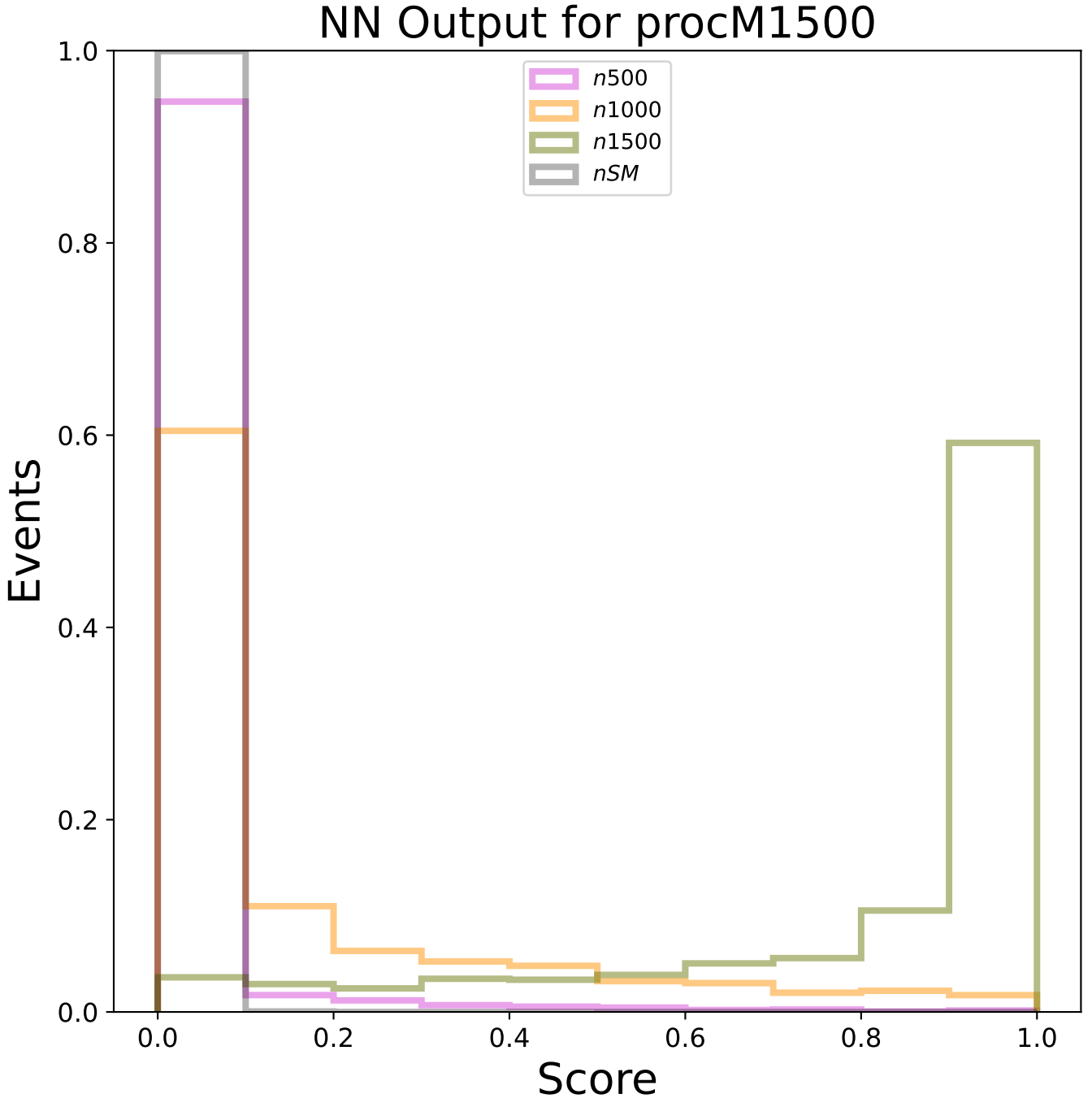
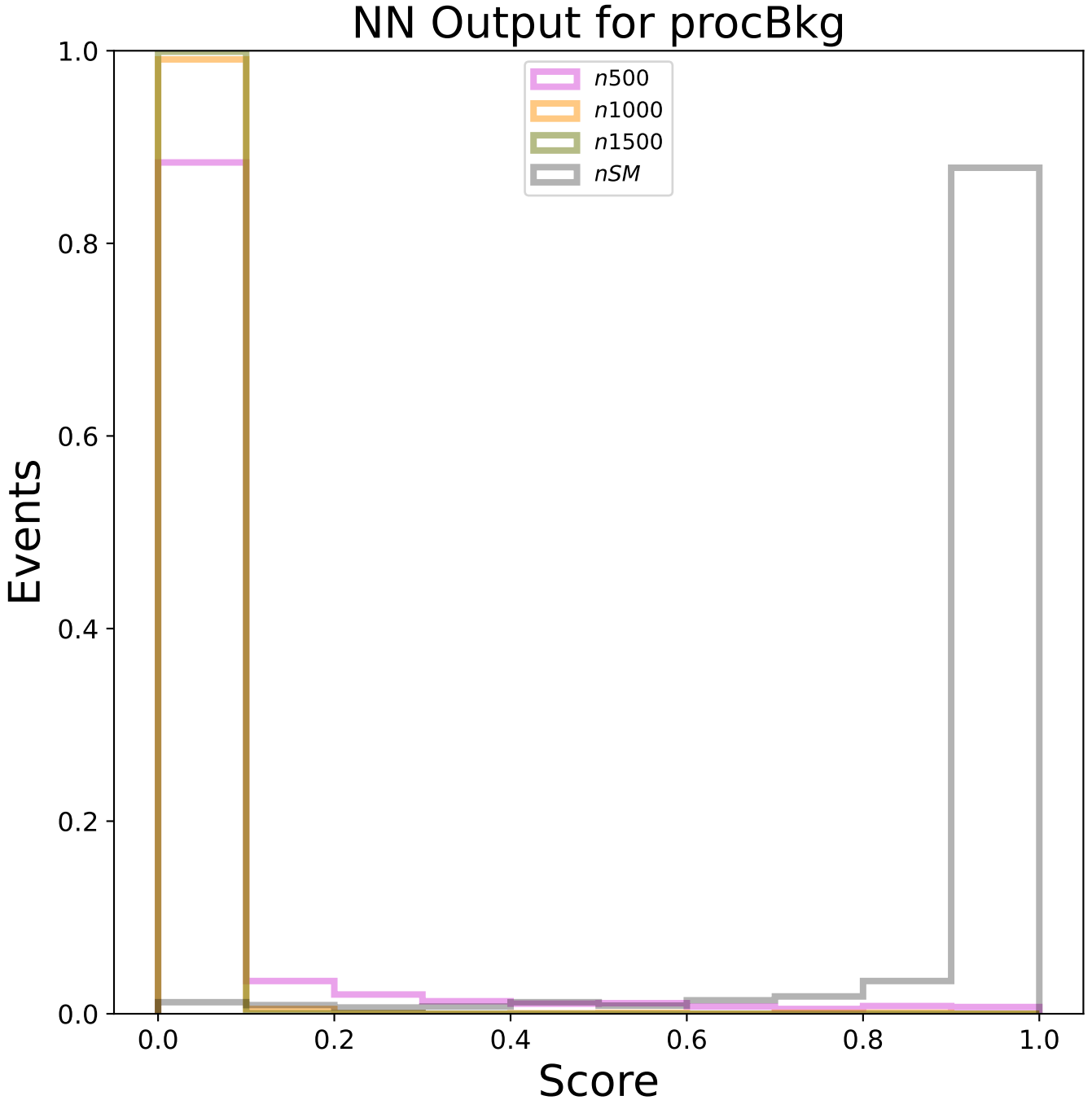
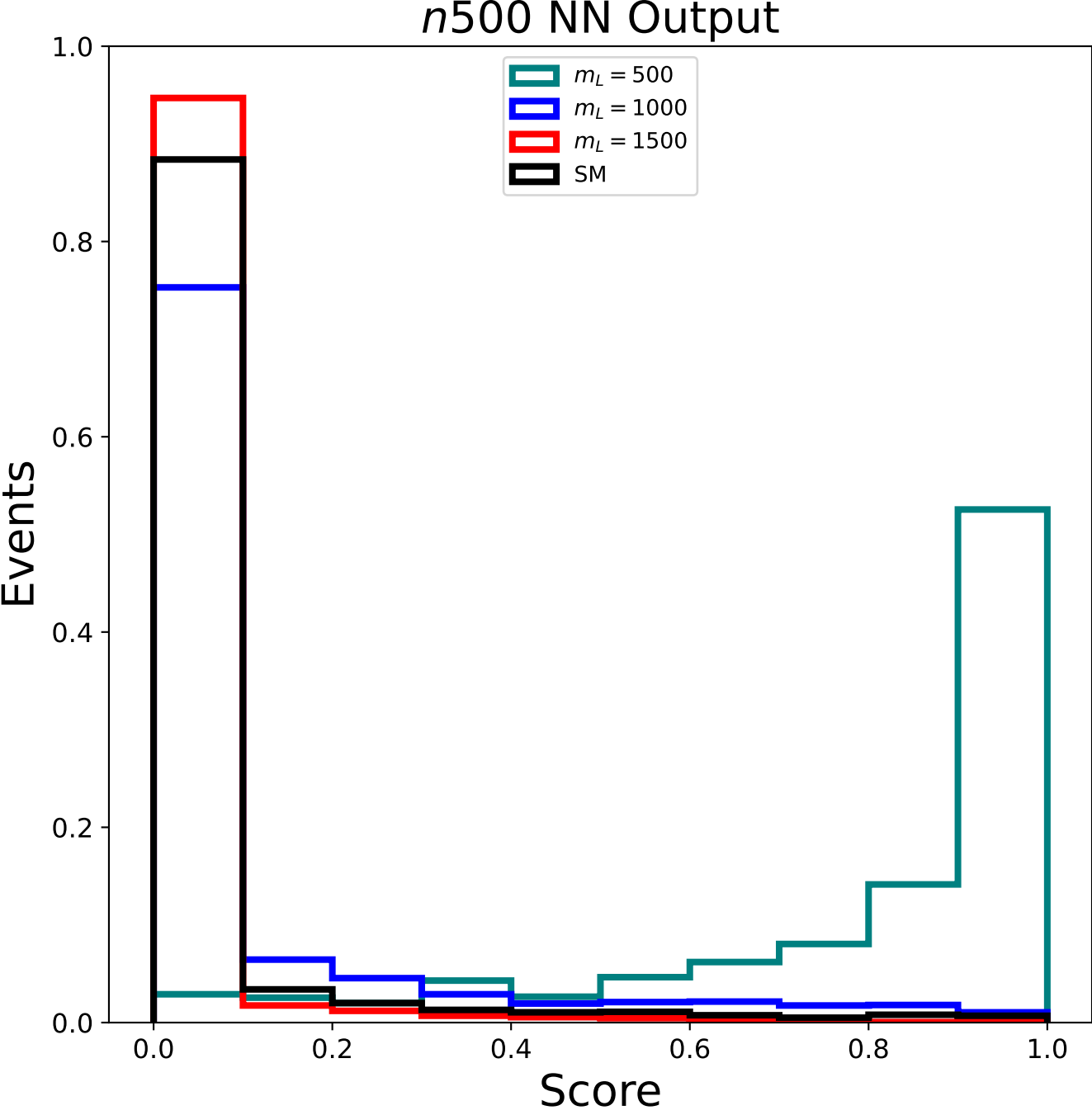

1.0
0.8
0.6
0.4
0.2
0.0
Signal Efficiency
0.0
0.2
0.4
0.6
0.8
1.0
Background Efficiency
n500 Training (AUC = 0.979)
n500 Validation (AUC = 0.976)
n1000 Training (AUC = 0.956)
n1000 Validation (AUC = 0.948)
n1500 Training (AUC = 0.977)
n1500 Validation (AUC = 0.975)
nSM Training (AUC = 0.997)
nSM Validation (AUC = 0.997)

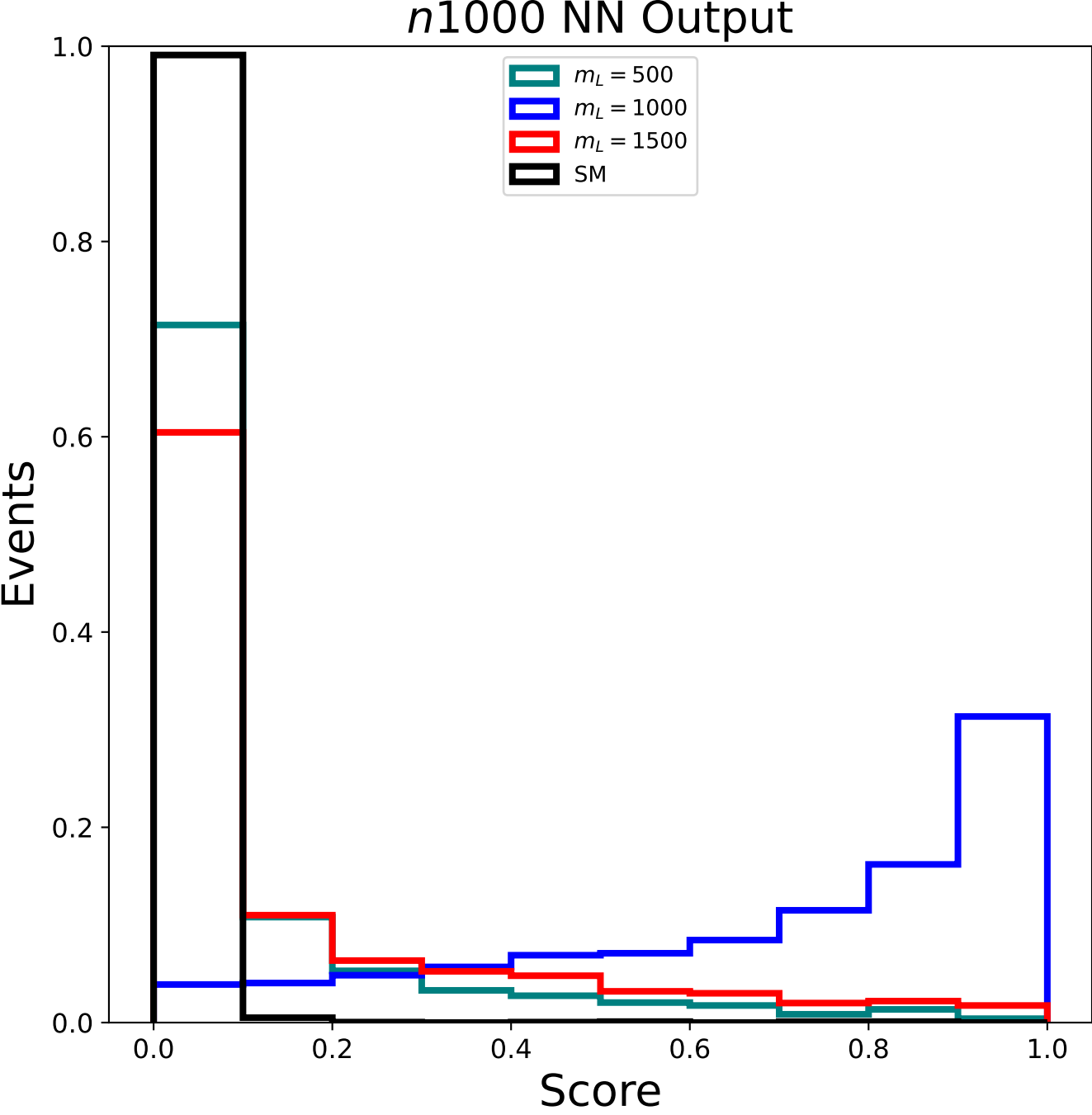

NN Output for procM500
n500
n1000
n1500
nSM
Events
Score
0.0
0.2
0.4
0.6
0.8
1.0

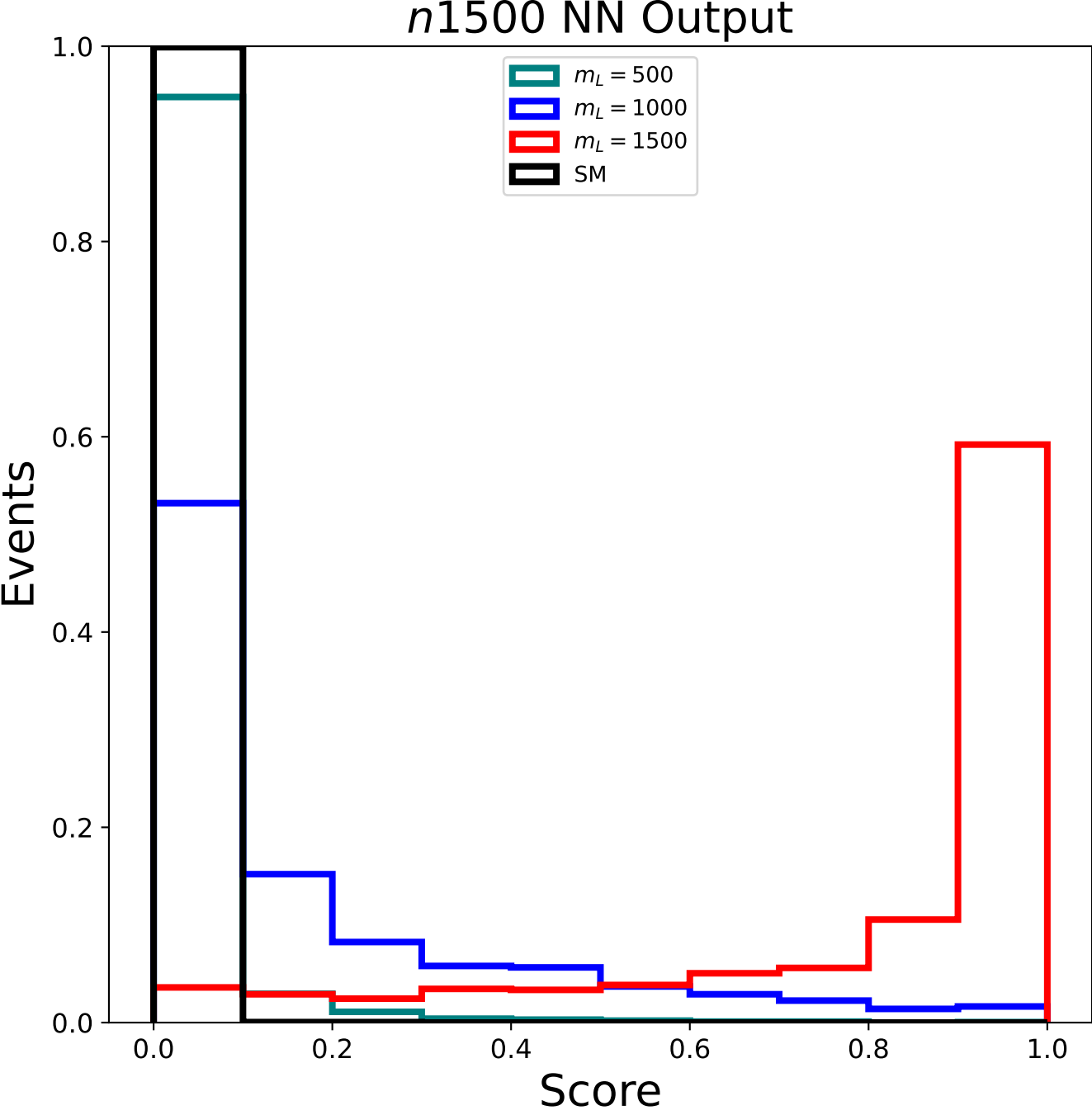

NN Output for procM1000
n500
n1000
n1500
nSM
Events
Score
0.0
0.2
0.4
0.6
0.8
1.0

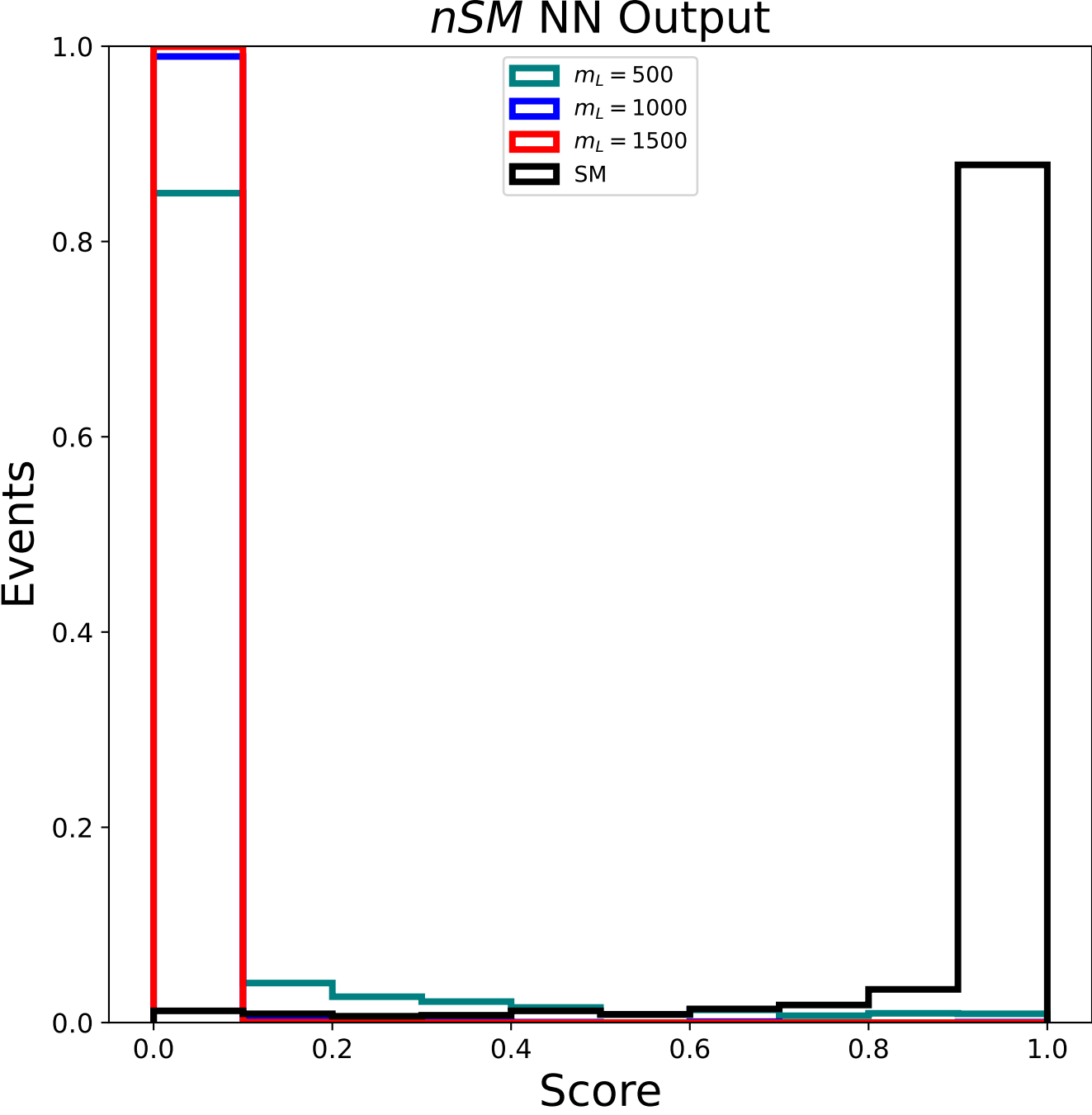

NN Output for procM1500
n500
n1000
n1500
nSM
Events
Score
0.0
0.2
0.4
0.6
0.8
1.0

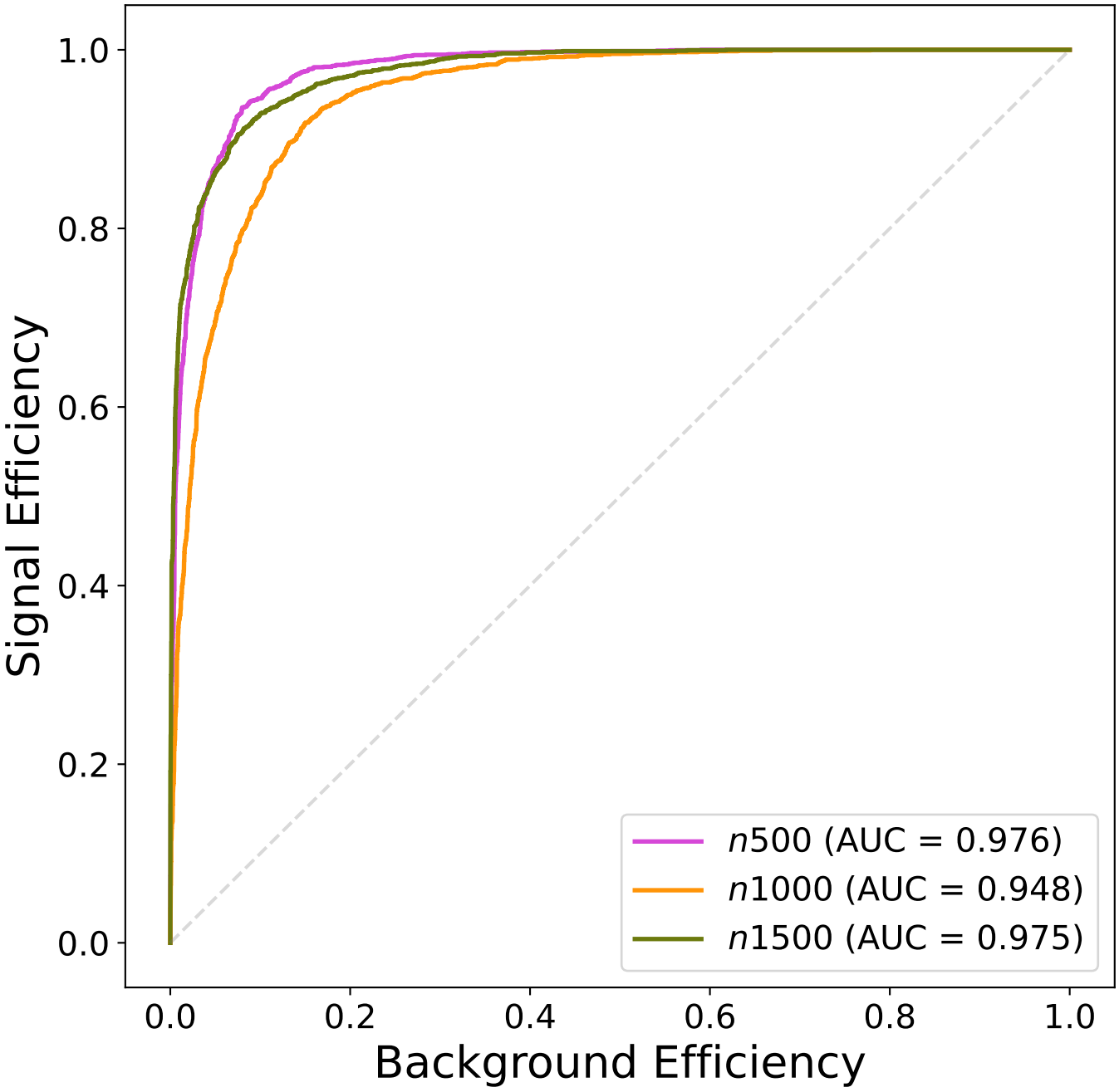

NN Output for procBkg
n500
n1000
n1500
nSM
Events
Score
0.0
0.2
0.4
0.6
0.8
1.0


$n500$ NN Output
$m_L = 500$
$m_L = 1000$
$m_L = 1500$
SM
Events
Score
0.0
0.2
0.4
0.6
0.8
1.0


$n1000$ NN Output
$m_L = 500$
$m_L = 1000$
$m_L = 1500$
SM
Events
Score
0.0
0.2
0.4
0.6
0.8
1.0


$n1500$ NN Output
$m_L = 500$
$m_L = 1000$
$m_L = 1500$
SM
Events
Score
0.0
0.2
0.4
0.6
0.8
1.0


nSM NN Output
$m_L = 500$
$m_L = 1000$
$m_L = 1500$
SM
Events
Score
0.0
0.2
0.4
0.6
0.8
1.0


1.0
0.8
0.6
0.4
0.2
0.0
Signal Efficiency
0.0
0.2
0.4
0.6
0.8
1.0
Background Efficiency
n500 (AUC = 0.976)
n1000 (AUC = 0.948)
n1500 (AUC = 0.975)